IAR University

Institute of Advanced Research

The University for Innovation

Question Bank (March 2026)	
Program Name: B.Tech CE	
Course Code: MIT104T	Semester:IV
Course Title: Operating System	Submission Date:- Before 3/3/2026
Credits: 3	Unit- 1,2,3

Note: Write any 5 questions of 2 marks and write any 5 questions of 5 marks.

Sr. No.	Questions	Marks
Q-1.	The Process Control Block (PCB) stores information such as process state, program counter, and CPU registers. (True / False)	1
Q-2.	Which of the following is a real-time scheduling algorithm? a) FCFS b) SJF c) Round Robin d) EDF	1
Q-3.	The operating system service that handles communication between processes is called _____.	1
Q-4.	A context switch occurs when the CPU switches from executing one process to another. (True / False)	1
Q-5.	A race condition occurs when multiple processes execute concurrently and access shared data. (True / False)	1
Q-6.	Which operating system service is responsible for creating and terminating processes? (A) File management (B) Memory management	1

	(C) Process management (D) Device management	
Q-7.	Semaphores are used to solve synchronization problems in concurrent processes. (True / False)	1
Q-8.	A microkernel architecture places most OS services in kernel space. (True / False)	1
Q-9.	Which of the following scheduling algorithms is non-preemptive by default? (A) Round Robin (B) Shortest Job First (C) Priority Scheduling (D) Rate Monotonic	1
Q-10.	In a _____ operating system structure, all system services execute in kernel mode.	1
Q-11.	What is a system call? Give any two examples of system calls.	2
Q-12.	Define an Operating System and mention any two services provided by it.	2
Q-13.	Explain the purpose of a Process Control Block (PCB) and list any two fields stored in it.	2
Q-14.	Define CPU scheduling. State any two scheduling objectives.	2
Q-15.	What is a thread? Mention any two benefits of multithreading.	2
Q-16.	What is a race condition? State one situation where it may occur.	2
Q-17.	Define turnaround time and waiting time in CPU scheduling.	2
Q-18.	Define a critical section and explain the need for mutual exclusion.	2
Q-19.	Write the pseudocode of Peterson's Solution for two processes P0 and P1.	2
Q-20.	Differentiate between Micro kernel and Monolithic kernel scheduling.	2

<table>
<tr><td>Q-21.</td><td>Differentiate between a process and a thread.</td><td>2</td></tr>
<tr><td>Q-22.</td><td>Apply the Pre-emptive Shortest Job First (Shortest Remaining Time First –
SRTF) scheduling algorithm and calculate: Waiting Time, Turnaround Time,
Average Waiting Time, Average Turnaround Time.

<table><tr><th>Process</th><th>Arrival Time (AT)</th><th>Burst Time (BT)</th></tr><tr><td>P1</td><td>0</td><td>7</td></tr><tr><td>P2</td><td>2</td><td>4</td></tr><tr><td>P3</td><td>1</td><td>2</td></tr><tr><td>P4</td><td>3</td><td>1</td></tr><tr><td>P5</td><td>4</td><td>3</td></tr></table></td><td>5</td></tr>
<tr><td>Q-23.</td><td>Explain semaphores and illustrate how they solve the Producer–Consumer problem.</td><td>5</td></tr>
<tr><td>Q-24.</td><td>Describe the different states of a process and explain the process state transitions with a neat diagram.</td><td>5</td></tr>
<tr><td>Q-25.</td><td>Describe the Reader - Writer problem in detail with pseudocode.</td><td>5</td></tr>
<tr><td>Q-26.</td><td>Describe CPU scheduling criteria and explain throughput, turnaround time, waiting time, and response time.</td><td>5</td></tr>
<tr><td>Q-27.</td><td>Apply the Round Robin scheduling algorithm with Quantum time = 3ms and calculate: Waiting Time, Turnaround Time, Average Waiting Time, Average Turnaround Time.</td><td>5</td></tr>
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	<table><tr><th>Process</th><th>Arrival Time (AT)</th><th>Burst Time (BT)</th></tr><tr><td>P1</td><td>0</td><td>6</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>5</td></tr><tr><td>P4</td><td>3</td><td>3</td></tr><tr><td>P5</td><td>4</td><td>7</td></tr></table>	
~~Q-28~~.	Describe the types of Operating Systems such as Batch, Multiprogramming, Time Sharing, and Real-Time Operating Systems with suitable examples.	5
~~Q-29~~	Explain Inter-Process Communication (IPC) mechanisms. Compare shared memory and message passing approaches.	5
Q-30.	Describe the Dining Philosophers problem and explain how it illustrates issues of synchronization and deadlock.	5